🚀 1-Click Fast Start: Pura App & Server Kaise Chalu Karein

👋 **Welcome Customer! Yeh Guide Kiske Liye Hai?**
Agar aapke paas poora project package aa gaya hai aur aap bina kisi technical jhanjhat ke turant **App, Server aur Admin Website** ko test karna chahte hain, toh yeh 5-page ki simple guide aapke liye hai.

1. Sabse Aasan Tarika: Batch (.bat) Files Se 1-Click Me Run Karna

Aapko koi command yaad rakhne ki zaroorat nahi hai. Project folder ke andar ready-made automation scripts hain:

🟢 Step 1: Server & Admin Website Chalu Karein

Folder me ja kar is file par **Double Click** karein:

```
START_ALL.bat
```

Yeh do windows open karega:
1. **Backend Relay API:** Port 4000 par start ho jayegi.
2. **Admin Web Panel:** Port 5173 par start ho jayega.

📱 Step 2: Phone Me App Install Karein

Phone ko USB cable se jodein aur double click karein:

```
INSTALL_ANDROID_USB.bat
```

Yeh script connected phone ko automatically detect karke ready APK install karegi aur phone screen par app turant open kar degi!

2. Browser Me Kaunsa URL Kholna Hai?

Service	URL Link	Screen Par Kya Dikhega?
Admin Web Portal	`http://localhost:5173`	Real-time Diagnostic Dashboard, Server Health & Logs.
Backend Health Check	`http://localhost:4000/health`	Green JSON status: `{"status":"UP"}` dikhega.
Live Cloud Backup	`https://smart-expenses-kuob.onrender.com`	Cloud deployed API endpoint status.

3. Kaam Khatam Hone Par Server Band Kaise Karein?

🔴 **Hamesha STOP_ALL.bat Use Karein:**
Direct window cross (X) mat karein. Folder me di gayi `STOP_ALL.bat` par double-click karein. Yeh background ports (4000 aur 5173) ko smoothly band kar deta hai taaki agli baar "*Port in use*" ka koi error na aaye.

📂 GitHub & Folder Structure: Kidar Kya Hai?

🌐 **Official GitHub Repository:**
Project ka official source code link: https://github.com/smartexpensehelp-sketch
Customer delivery archive zip: `SmartExpenses-Complete-Project.zip`

1. Project Folder Ka Map (Har Folder Ka Kaam)

Jab aap ZIP file extract karte hain, toh aapko yeh 4 main folders aur automation files milti hain:

Folder / File Name	Category	Iske Andar Kya Hai Aur Iska Use Kya Hai?
`android/`	Mobile App	Pure Java 17 aur XML layouts ka complete Android Studio project. Room SQLite local database, voice engine aur UI screens isi ke andar hain.
`backend/`	Relay Server	Node.js, Express aur TypeScript ka backend API server. Mobile app aur Supabase database ke beech bridge ka kaam karta hai (Port 4000).
`admin/`	Web Portal	React 18 aur Vite ka Admin Dashboard source code. Isme system diagnostics, database connection status aur network telemetry monitor hoti hai.
`APK/`	Ready Binary	Pre-compiled `SmartExpenses-v1.0.apk` file. Agar aapko Android Studio open nahi karna hai, toh aap is APK ko direct phone me install kar sakte hain.
`*.bat files`	Automation	`START_ALL.bat`, `STOP_ALL.bat`, `INSTALL_ANDROID_USB.bat` aur `RUN_AND_INSTALL.bat`.

2. GitHub Par Important Files Kidar Hain?

Android Ke Main Java Codes

- **Room DB Model:** `android/app/src/main/java/.../entity/TransactionEntity.java`
- **Database Queries:** `.../dao/TransactionDao.java`
- **Voice AI Sheet:** `.../ui/voice/VoiceBottomSheet.java`
- **Background Sync:** `.../worker/SyncWorker.java`

Backend & Server Files

- **Main Server:** `backend/src/server.ts`
- **Sync Push/Pull:** `backend/src/routes/sync.ts`
- **Admin Telemetry:** `backend/src/routes/monitoring.ts`
- **Environment Keys:** `backend/.env`

📌 **Rule Jo Yaad Rakhna Hai:** Android project me ek bhi Kotlin file (`.kt`) nahi hai! Poora code **Pure Java & XML** me likha gaya hai.

Supabase Cloud Database & Backend Setup

1. Supabase Kya Hai Aur Yeh Kyu Use Kiya Hai?

Supabase ek secure cloud platform hai jo background me authentic **Enterprise PostgreSQL 15 Database** run karta hai. Jab phone me internet hota hai, toh app ka sara financial data encrypt hokar Supabase me permanently safe ho jata hai.

Row Level Security (RLS)

Har user ka data doosre user se 100% isolated rehta hai. User A kabhi User B ka transaction na padh sakta hai na badal sakta hai.

Paise Integer Precision

Supabase table me amount `amount_minor BIGINT` (Paise) me store hota hai, jisse ₹1 paise ka bhi rounding error nahi aata.

2. Supabase Dashboard Me Live Data Kaise Check Karein?

1. Apne browser me **https://supabase.com/dashboard** par jayein aur account login karein.
2. Apna Project select karein (Project Ref ID: `dymorpzwrmvfpubgsgqb`).
3. Left side menu me **Table Editor** par click karein.
4. Aapko tables dikhayi dengi:
   - `transactions`: Phone se sync hue kharche aur aamadni ki live rows.
   - `profiles`: Registered users ki email aur details.
   - `accounts`: Wallets (Cash, Bank Account, Savings) aur unka minor balance.

3. Backend .env File (Keys Kaha Hoti Hain?)

Backend folder me `backend/.env` file hoti hai jisme cloud keys pre-configured hain:

```
PORT=4000
NODE_ENV=development
SUPABASE_URL=https://dymorpzwrmvfpubgsgqb.supabase.co
SUPABASE_ANON_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
SUPABASE_SERVICE_ROLE_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
JWT_SECRET=super_secret_jwt_hs256_key_for_smart_tracker
ADMIN_API_KEY=admin_secret_key_2026
```

✅ **Customer Advantage:** Aapko yeh keys manually bharne ki koi zaroorat nahi hai. Humne sab kuch pehle se link aur test karke package kiya hai. Sirf `START_ALL.bat` dabate hi backend automatically database se connect ho jata hai!

🛠️ Android Studio Me Project Kaise Open Karein

⚠️ **Sabse Badi Common Galati Jo Beginners Karte Hain:**
Android Studio me poora root folder open nahi karna hota! Agar aap root folder select karenge toh Gradle detect nahi hoga. Hamesha sirf `android` folder ko select karein!

1. Step-by-Step Android Studio Open Karne Ka Tarika

STEP 1 **Android Studio Launch Karein**
Apne PC me Android Studio open karein. Welcome screen par **"Open"** (ya **File** → **Open**) par click karein.

STEP 2 **Sahi Folder Select Karein**
Folder browse karke yaha jayein:
`...\divya\android` ← *(Sirf is 'android' folder par click karke OK dabayein).*

STEP 3 **Gradle Sync Hone Dein (1-2 Minute)**
Project khulte hi bottom right me *"Gradle build running..."* dikhega. Internet on rakhein taaki libraries verify ho sakein. Jab neeche **"Gradle sync finished"** green tick aa jaye, tab project compile ke liye ready hai.

2. JDK 17 Configuration Check Karein

Agar kabhi Java version error aaye toh yeh 10-second setting check karein:

1. Android Studio me **File** → **Settings** (Mac par Preferences) open karein.
2. **Build, Execution, Deployment** → **Build Tools** → **Gradle** par jayein.
3. **Gradle JDK** me select karein: **"Embedded JDK 17"** ya **"Android Studio java runtime (jbr)"**.
4. Apply aur OK dabayein.

3. App Run (Compile) Kaise Karein?

Phone / Emulator Select Karein

Top toolbar me device dropdown dekhein. Apna connected real phone (jaise **Nova 1 5G**) ya Android Emulator select karein.

Green Play Button Dabayein

Toolbar me green color ke **Play Button (Run 'app')** → `Shift + F10` par click karein. App compile hokar phone par khul jayega!

📱 Real Phone Me USB Debugging & Network Setup

Customer ya teacher ke samne physical phone par app run karke dikhana sabse convincing hota hai. Kisi bhi Android phone (jaise **Nova 1 5G**) ko connect karne ka full step-by-step method:

1. Phone Me Developer Mode & USB Debugging On Karna

STEP A **Developer Options Unlock Karein**
Phone ki **Settings** → **About Phone** me jayein. **Build Number** (kuch phones me Version) par lagataar **7 baar tap karein**. Message aayega: "*You are now a developer!*".

STEP B **USB Debugging On Karein**
Settings me wapas ja kar **Additional Settings** → **Developer Options** me jayein. Waha **"USB Debugging"** ka toggle ON karein.

STEP C **Computer Ko Authorize Karein**
Phone ko USB cable se PC se connect karein. Phone screen par ek prompt aayega: "*Allow USB Debugging from this computer?*" **"Always allow from this computer"** par tick mark lagayein aur **OK** dabayein.

2. 1-Click Phone Me Install Karna (Script)

Developer Mode on karne ke baad aapko kisi command ki zaroorat nahi hai. Simply double-click karein:

```
INSTALL_ANDROID_USB.bat
```

Yeh script ADB ke through aapke phone me direct APK push karegi aur screen par automatic welcome splash screen khol degi!

3. USB Tethering / Wi-Fi Par Local Server Connect Karna

🌐 **Agar Phone Aur PC Dono Ek Hi Wi-Fi Par Hain:**
PC ka Local IP address dekhein (CMD me `ipconfig` type karein, e.g. `192.168.1.15`).
Phone me app chalate waqt backend URL automatically Render Cloud API (`https://smart-expenses-kuob.onrender.com`) par sync hota hai, isliye agar aap local IP configure na bhi karein tab bhi cloud sync 100% seamlessly kaam karta hai!

🎉 **Badhai Ho! Customer Setup Poora Ho Gaya:**
Aapka Smart Expenses Tracker ab physical phone par live chal raha hai. Aap offline transactions add kar sakte hain, voice se bolkar kharcha save kar sakte hain, aur website dashboard par live health monitor kar sakte hain!